Viscosity of Liquids

Pre-Lab Plan
Unit Operations Lab 2
September 19th, 2019
Group 1, Thursday, PM section

Alexis Miranda
Dulce Colindres
Kathya Chavez

1. Experimental Objective

In this experiment, viscosity of a fluid is calculated using the measured pressure drop and flow rate across a capillary tube using the Hagen-Poiseuille equation for a Newtonian fluid.

2. Experimental

2.1 Apparatus

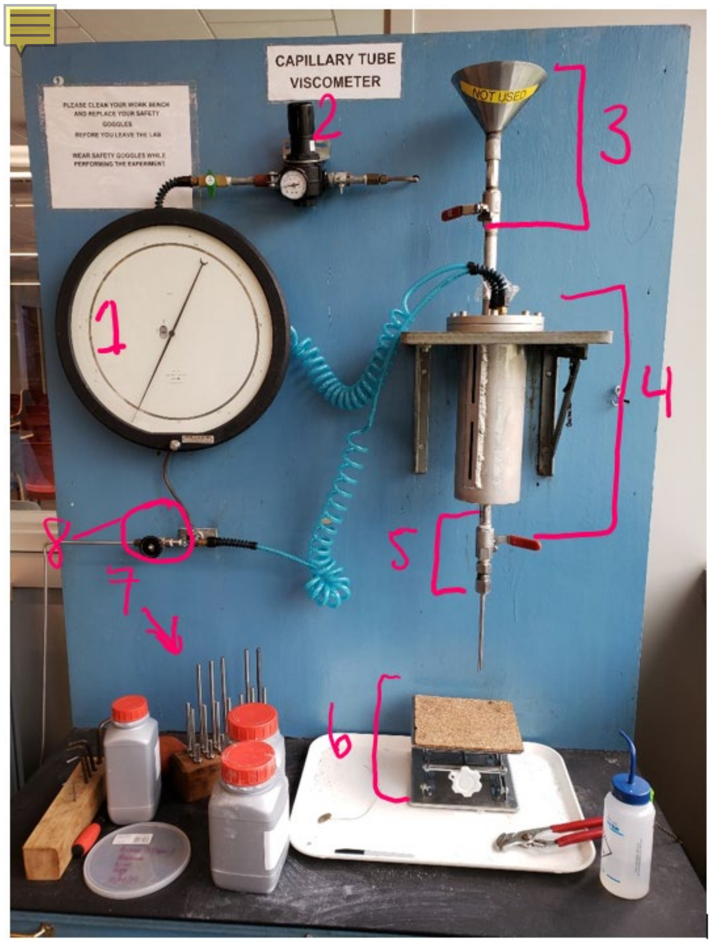

Figure 1 : Apparatus for Capillary Tube Viscometer Lab

Table 1 : Apparatus Components

Component #	Function
1 - Pressure meter	Shows how much pressure is present in the vessel.
2 - Pressure changer	When turned will increase or decrease pressure in vessel .
3 - Funnel and valve	Where the fluid is added to get to the vessel . It is not used
4 -Vessel	Contains fluid for which the viscosity is to be measured. It is pressurized with air
5 -Outlet valve	When opened , it will release the fluid from the vessel.
6 - Standing plate	Where we place the jar to collect the fluid leaving the vessel
7 - Capillary tubes (of different length : diameter ratio)	Where non-newtonian fluid passes, each will give different flow rates
8 - Vent valve	It is opened to decrease pressure in vessel, closed to increase pressure in vessel

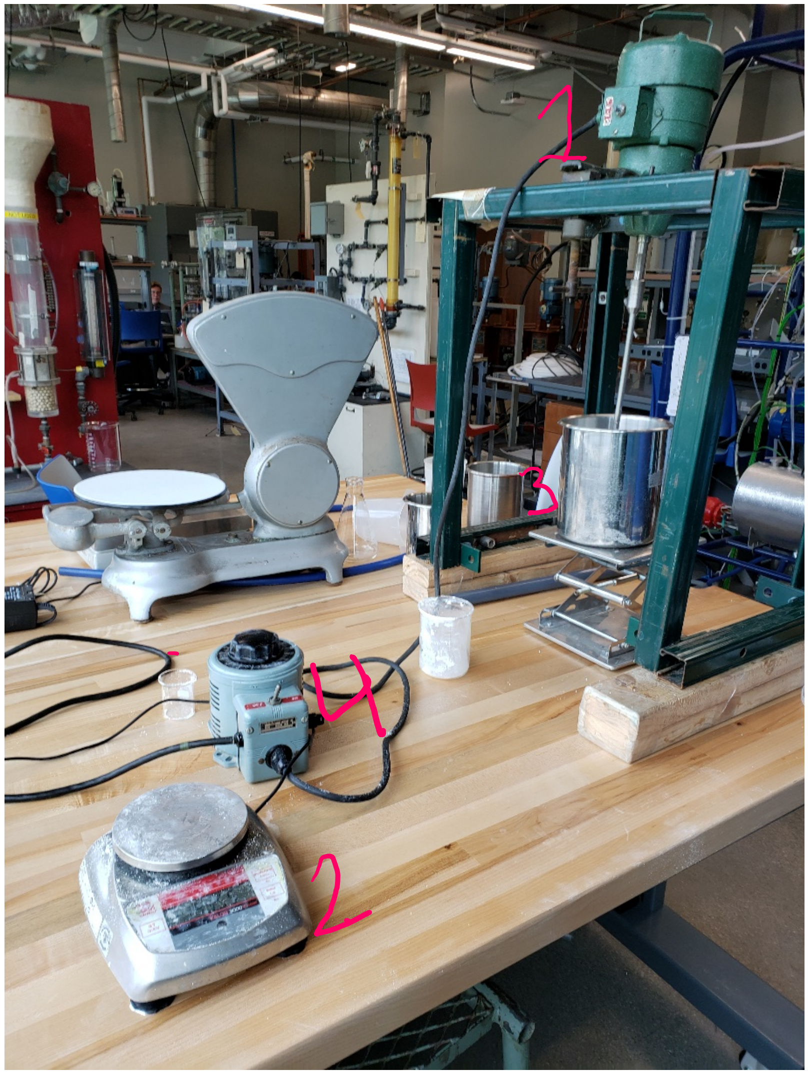

Figure 2 : Mixer, balance, and bucket used to mix non-newtonian fluid

Component #	**Function**
1 - Mixer	Mixing tool to mix materials
2 - Balance	To measure dry material
3 – Bucket with non-Newtonian fluid	Containing all experimental fluid
4- Speed controller	Controls rpms for Mixer

2.2 Materials and Supplies

Materials	Usage
Warm Tap Water	Mixed with CARBOPOL to make a non-newtonian fluid.
CARBOPOL	Used to make the non-newtonian fluid
Power Mixer	Mixes water and CARBOPOL to make the non newtonian fluid
Stainless steel bucket	Contains water and CARBOPOL for mixing using power mixer
Nitrile Gloves	Protect hand from contacting CARBOPOL solution
Lab Coat	Protect from any fluid splashes.
Weight	Measures mass of the mixed fluid for density calculation
Air	Used to pressurize the vessel
Pincers	Used to open the screws on the vessel
Paper towels	Used to clean any excess 8% CARBOPOL fluid
Beakers or cylinders	Used to contain fluid or water. Also used for volume measurement of non newtonian fluid.

2.3 Experimental Plan

Begin by preparing a non-newtonian fluid for experiment. A non-newtonian fluid can be created by mixing 0.8 wt% solution of CARBOPOL, which is a powder polymer used as a thickening agent, with warm tap water. A power mixer is available to properly mix both the CARBOPOL and water together. After creating the needed fluid, begin by filling a capillary tube and placing it in the apparatus. The tube will go under pressurization during the experiment, so several precautions must be taken. When dispersing the fluid in the reservoir, remember to open the vent valve, which will depressurize the vessel. Also, remember to close the vent valve before pressurizing to begin the experiment. The capillary tubes come in different diameters in length. When using a certain capillary tube, measure the flow rate with several different pressure differentials. Repeat with a capillary tube that is a different size and/or diameter. Our plan is to test 2 different capillary diameters at 3 different lengths as well as each trial at 3 different pressures. This means we will have to record and produce 18 viscosity values in total.

We will evaluate the volumetric flow and graph it against the recorded pressure drops for each run. This will produce a straight line of which we can calculate viscosity from the slope. The volumetric flow rate can be found using the Ostwald-deWaele viscosity model.

Equations

Weight percent= $\frac{Weight\ of\ solute}{weight\ of\ solution} * 100$

0.8wt%=$\frac{m_p}{m_w} * 100$
Basis is 2L of water, density of water is 1000g/L
M_p = 16.00g: Mass of powder used
M_w= 2L: Mass of water used

To create 0.8wt% solution we added 16g of Mix to 2L of water, then adding nNaOH

Independent & Dependent Variables

Independent	**Notes**
Pressure	Pressure is added or removed from the vessel, which would affect the flow rates of fluid, thus provides data for the viscosity of fluid
Diameter/length of capillary tube	We change between different length and diameter of capillary tubes , for a total of 9 combinations , to measure flow rates of the non-newtonian fluid.

Dependent	**Notes**
Flow Rates	When changing pressure or length/diameter of capillary tube, flow rates will fluctuate

Data Collection

Table 3 : Flow Rates of non-newtonian fluid with varying capillary tube sizes and pressures (gram per second)

	L = 3 cm		L = 6 cm		L = 9 cm	
∆P (psig)	R = 0.062in	R = 0.053in	R = 0.062in	R = 0.053in	R = 0.062in	R = 0.053in
20						
30						
40						

- This table will be filled out during experiment, thus presented in final report.

Project Safety Evaluation

Project Title: *Viscosity of Fluids* *Date: 09/19/2019*

	Yes/No	
Potential electrical hazards?	*Yes*	*Control water spills around Mixer speed controller.*
Potential mechanical hazards?	*Yes*	*Power mixer should be handled carefully, and pinch points.*
Potential pressure hazards?	*Yes*	*Be sure to Depressurization system when changing capillary tube. Safety pressure release valve found behind the system can be used in case of emergency.*
Potential temperature hazards?	*No*	*Experiment will be conducted at room temperature.*
Hazardous/toxic chemicals involved?	*Yes*	*CARBOPOL should not be inhaled or come in contact with skin. For the former, seek medical attention. For the latter, wash skin in contact thoroughly* *NaOH is a high basic, Corrosive inorganic Base; avoid contact and ingestion.*
Gloves required?	*Yes*	*Type : Nitrile*
MSDS reviewed by all team members?	*No*	*MSDS for CARBOPOL 940 was reviewed*
Process Parameters	***Max Expected***	***List location and possible hazards and precautions***
Temperature (ºC)	*Room Temp*	*Experiment will be conducted at room temperature*
Pressure (psia)	*60 psig*	*Experiment operates at or below max of 60 psig. Depressurize system when changing any parts to the experiment.*
Safety Controls	***yes/no***	***If yes, Provide description***
Use of personal protective equipment (PPE) *Elimination*	*Yes*	*Wearing safety glass, coat to avoid chemical splash, nitrile to prevent skin contact with either CARBOPOL and non-newtonian fluid.*

Emergency pressure release	*yes*	*Behind testing area there is a pressure release valve for any experimental emergency to depressurize system.*

*I have read relevant background material for the Unit Operations Laboratory entitled: "**Non-Newtonian Viscosity Lab**" and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.*

Alexis Miranda — *9/19/19*
signature of team member — *Date*

Dulce Colindres — *9/19/19*
signature of team member — *Date*

Kathya Chavez — *9/19/19*
signature of team member — *Date*